Definition 1 (LRA(G)). Let $G \triangleq (N, T, P, S)$ be a context-free grammar.

(i) $G' := (N', T', P', S') \triangleq (N \uplus \{S'\}, T \uplus \{\$\}, P \cup \{[S' \to S\$]\}, S')$.

(ii) $V \triangleq N \uplus T$. $V' \triangleq N' \uplus T'$.

(iii) $\beta \underset{\mathbf{rm}}{\Longrightarrow}_{P'} \gamma \overset{\text{def}}{\Longleftrightarrow} (\beta, \gamma) \in \{(\alpha A z, \alpha\omega z) \mid [A \to \omega] \in P', \alpha \in V'^*, z \in T'^*\}$.

(iv) $I_{\text{LR}(0)} \triangleq \{[A \to \alpha \cdot \beta] \mid [A \to \alpha\beta] \in P'\}$.

(v) $\text{Cl}(q) =_\mu q \cup \{[A \to \varepsilon \cdot \omega] \mid [A \to \omega] \in P', [B \to \beta \cdot A\gamma] \in \text{Cl}(q)\}$ for $q \in \mathcal{P}(I_{\text{LR}(0)})$.

(vi) $q_0 := \text{Cl}(\{[S' \to \varepsilon \cdot S\$]\})$.

(vii) $\text{GOTO}(q, X) := \text{Cl}(\{[A \to \alpha X \cdot \beta] \mid [A \to \alpha \cdot X\beta] \in q\})$ for $q \in \mathcal{P}(I_{\text{LR}(0)})$ and $X \in V'$.

(viii) $Q \triangleq \text{PT} \setminus \{\emptyset\}$ where $\text{PT} =_\mu \{q_0\} \cup \{\text{GOTO}(q, X) \mid q \in \text{PT}, X \in V'\}$.

(ix) $\varepsilon : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \wedge p = q$. $X\alpha : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \wedge \alpha : \text{GOTO}(p, X) \to q$.

(x) $\mathbf{Config} \triangleq \{(\alpha : p \to q, z) \mid \alpha \in V'^*, p \in Q, q \in Q, z \in T'^*, \alpha : p \to q\}$.

(xi) $\delta(q, X) := \text{GOTO}(q, X)$ for $q \in Q$ and $X \in V'$.

(xii) $\mathbf{reduce}(q, t) := \{[A \to \omega] \mid [A \to \omega \cdot \varepsilon] \in q\}$ for $q \in Q$ and $t \in T'$.

(xiii) Let $\vdash$ be a binary relation on the set $\mathbf{Config}$ with two introduction rules:

$$\frac{q'' = \delta(q', t)}{(\alpha : q \to q', tz) \vdash (\alpha t : q \to q'', z)} \text{ Shift}$$

$$\frac{\alpha : q \to p \quad \omega : p \to q' \quad [A \to \omega] \in \mathbf{reduce}(q', t) \quad q'' = \delta(p, A)}{(\alpha\omega : q \to q', tz) \vdash (\alpha A : q \to q'', tz)} \text{ Reduce}(A \to \omega)$$

(xiv) $q_f := \delta(\delta(q_0, S), \$)$.

(xv) The language accepted by LRA(G) is defined as follows:

$$L(\text{LRA}(G)) := \{z \in T^* \mid (\varepsilon : q_0 \to q_0, z\$) \vdash^* (S\$: q_0 \to q_f, \varepsilon)\}.$$

Fact 2. *It is known that $L(G) = L(\text{LRA}(G))$, where $L(G) \triangleq \{z \in T^* \mid S \Rightarrow_P^* z\}$.*

Proof. For a configuration $c = (\alpha : p \to q, u) \in \mathbf{Config}$ write $\text{Y}(c) := \alpha u \in V'^*$ for its *yield*.

Viable-prefix lemma. For $\eta \in V'^*$:

(V1) $(\exists q \in Q)\ \eta : q_0 \to q$ iff η is a viable prefix of G';

(V2) if $\eta = \eta'\omega$ with $\eta' : q_0 \to p$, $\omega : p \to q$, and this occurrence of ω is the handle of $[A \to \omega] \in P'$ (i.e. $S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \eta' A y \underset{\mathbf{rm}}{\Longrightarrow}_{P'} \eta'\omega y$, $y \in T'^*$), then $[A \to \omega \cdot \varepsilon] \in q$, so $[A \to \omega] \in \mathbf{reduce}(q, t)$ for the first symbol t of y.

Both are the standard correctness of the characteristic LR(0) automaton and follow by induction on the clauses Cl, GOTO of Definition 1(v),(vii) and the path relation (ix).

Step invariant. For $c \vdash c'$,

$$\text{Y}(c') = \text{Y}(c) \ \ (\text{Shift}), \qquad \text{Y}(c') \underset{\mathbf{rm}}{\Longrightarrow}_{P'} \text{Y}(c) \ \ (\text{Reduce}). \tag{$*$}$$

Indeed a Shift $(\alpha : q \to q', tz) \vdash (\alpha t : q \to q'', z)$ has $\text{Y} = \alpha t z$ on both sides, and a Reduce $(\alpha\omega : q \to q', tz) \vdash (\alpha A : q \to q'', tz)$, $[A \to \omega] \in P'$, has $\text{Y}(c) = \alpha\omega t z$, $\text{Y}(c') = \alpha A t z$ with $\alpha \in V'^*$, $tz \in T'^*$, so $\alpha A t z \underset{\mathbf{rm}}{\Longrightarrow}_{P'} \alpha\omega t z$ by Definition 1(iii). Iterating $(*)$ along $c \vdash^* c'$,

$$c \vdash^* c' \implies \text{Y}(c') \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \text{Y}(c). \tag{$**$}$$

Soundness, $L(\text{LRA}(G)) \subseteq L(G)$. Let $z \in L(\text{LRA}(G))$, so $(\varepsilon : q_0 \to q_0, z\$) \vdash^* (S\$: q_0 \to q_f, \varepsilon)$. The endpoints have yields $z\$$ and $S\$$, so $(**)$ gives

$$S\$ \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} z\$.$$

Since S' occurs on no right-hand side of P', $[S' \to S\$]$ is applied at no step of a derivation from $S\$$; thus every step uses a production of P. As $\$ \in T' \setminus T$ is never rewritten and never matched by a production of P, it rides unchanged at the right end, and deleting it gives $S \Rightarrow_P^* z$, i.e. $z \in L(G)$.

Completeness, $L(G) \subseteq L(\mathrm{LRA}(G))$. Let $z \in L(G)$ and fix a rightmost derivation

$$S\$ = \gamma_0 \underset{\mathbf{rm}\ \ P'}{\Longrightarrow} \gamma_1 \underset{\mathbf{rm}\ \ P'}{\Longrightarrow} \cdots \underset{\mathbf{rm}\ \ P'}{\Longrightarrow} \gamma_N = z\$,$$

which exists because $S \Rightarrow_P^* z$. Write the k-th step as

$$\gamma_k = \alpha_k A_k y_k \underset{\mathbf{rm}\ \ P'}{\Longrightarrow} \alpha_k \omega_k y_k = \gamma_{k+1}, \qquad [A_k \to \omega_k] \in P',\ \alpha_k \in V'^*,\ y_k \in T'^*,$$

where y_k ends in \$, hence is nonempty with first symbol t_k. The suffix ω_k of the viable prefix $\alpha_k\omega_k$ is the handle of $[A_k \to \omega_k]$, so by (V2) there are $p_k, q_k \in Q$ with

$$\alpha_k : q_0 \to p_k, \quad \omega_k : p_k \to q_k, \quad [A_k \to \omega_k \cdot \varepsilon] \in q_k, \quad [A_k \to \omega_k] \in \mathbf{reduce}(q_k, t_k).$$

Read the derivation from $\gamma_N = z\$$ down to $\gamma_0 = S\$$: starting at $(\varepsilon : q_0 \to q_0, z\$)$, for $k = N-1, \ldots, 0$ Shift the terminals lying between the current stack and the handle ω_k (each legal by the Shift rule, as δ is defined along the viable prefix $\alpha_k\omega_k$) until the configuration is $(\alpha_k\omega_k : q_0 \to q_k, y_k)$, then apply Reduce($A_k \to \omega_k$), obtaining by the line above and Definition 1(xiii)

$$(\alpha_k\omega_k : q_0 \to q_k,\ y_k) \ \vdash\ (\alpha_k A_k : q_0 \to \delta(p_k, A_k),\ y_k),$$

the configuration associated with γ_k. Each iteration inverts one $\underset{\mathbf{rm}\ \ P'}{\Longrightarrow}$ step; after all N of them and a final Shift of the endmarker \$ the input is exhausted and the stack carries $S\$$:

$$(\varepsilon : q_0 \to q_0, z\$) \ \vdash^*\ (S\$: q_0 \to q_f, \varepsilon), \qquad q_f = \delta(\delta(q_0, S), \$),$$

so $z \in L(\mathrm{LRA}(G))$. With soundness, $L(G) = L(\mathrm{LRA}(G))$. ☐

Theorem 3. *Define* $\mathbf{LA} : \{(q, [A \to \omega]) \mid q \in Q, [A \to \omega \cdot \varepsilon] \in q\} \to \mathcal{P}(T')$ *by*

$$\mathbf{LA}(q, [A \to \omega]) := \left\{ t \in T' \;\middle|\; S' \underset{\mathbf{rm}\ \ P'}{\overset{*}{\Longrightarrow}} \alpha A t z, \alpha\omega : q_0 \to q, \alpha \in V'^*, z \in T'^* \right\}. \tag{1}$$

Then, overriding $\mathbf{reduce} : Q \times T' \to \mathcal{P}(P')$ *of the* LR(0) *parser* LRA(G) *with*

$$(q, t) \mapsto \{[A \to \omega] \mid [A \to \omega \cdot \varepsilon] \in q, t \in \mathbf{LA}(q, [A \to \omega])\}$$

yields an LALR(1) *parser if there are no conflicts, which accepts the same language.*

Proof. Let $\vdash_{\mathbf{LA}}$ be $\vdash$ with **reduce** replaced by

$$\mathbf{reduce}_{\mathbf{LA}}(q, t) := \{[A \to \omega] \mid [A \to \omega \cdot \varepsilon] \in q,\ t \in \mathbf{LA}(q, [A \to \omega])\},$$

and let $L(\vdash_{\mathbf{LA}})$ be the language it accepts.

(⊆) Since $\mathbf{reduce}_{\mathbf{LA}}(q, t) \subseteq \mathbf{reduce}(q, t)$ for every (q, t), each Reduce instance of $\vdash_{\mathbf{LA}}$ is one of $\vdash$, and the Shift rule is shared; hence $\vdash_{\mathbf{LA}} \subseteq \vdash$, $\vdash_{\mathbf{LA}}^* \subseteq \vdash^*$, and

$$L(\vdash_{\mathbf{LA}}) \subseteq L(\mathrm{LRA}(G)) \overset{\text{Fact 2}}{=} L(G).$$

(⊇) Let $z \in L(G)$. The completeness construction of Fact 2 yields a computation

$$(\varepsilon : q_0 \to q_0, z\$) \ \vdash^*\ (S\$: q_0 \to q_f, \varepsilon)$$

each of whose Reduce steps inverts a rightmost step $\alpha A t z' \underset{\mathbf{rm}\ \ P'}{\Longrightarrow} \alpha\omega t z'$ and is taken at a configuration $(\alpha\omega : q_0 \to q, tz')$ with $[A \to \omega \cdot \varepsilon] \in q$ and $\alpha\omega : q_0 \to q$. As $S' \underset{\mathbf{rm}\ \ P'}{\Longrightarrow} S\$ \underset{\mathbf{rm}\ \ P'}{\overset{*}{\Longrightarrow}} \alpha A t z'$, the witnesses α, z' and the path $\alpha\omega : q_0 \to q$ meet the defining condition (1) of **LA**, so

$$t \in \mathbf{LA}(q, [A \to \omega]), \qquad \text{i.e.} \qquad [A \to \omega] \in \mathbf{reduce}_{\mathbf{LA}}(q, t).$$

Thus every step is also a $\vdash_{\mathbf{LA}}$-step and the same computation accepts z; hence $L(G) \subseteq L(\vdash_{\mathbf{LA}})$. Combining, $L(\vdash_{\mathbf{LA}}) = L(G)$.

For the table, each pair (q, t) carries the shift to $\delta(q, t)$ when $\delta(q, t) \neq \emptyset$ and the reduces $[A \to \omega]$ with $[A \to \omega \cdot \varepsilon] \in q$ and $t \in \mathbf{LA}(q, [A \to \omega])$ — exactly the LR(0) cores annotated with the LALR(1) lookahead sets **LA**. If no (q, t) enables two distinct actions — no shift/reduce ($\delta(q, t) \neq \emptyset$ together with an enabled reduce) and no reduce/reduce (two distinct enabled reduces) — then $\vdash_{\mathbf{LA}}$ is deterministic with one symbol of lookahead, i.e. an LALR(1) parser. ☐

Theorem 4. *Letting $R\,!\,x \triangleq \{y \mid (x,y) \in R\}$, define relations* **Read** *and* **Follow** *from the set*

$$\{(p,A) \mid p \in Q, A \in N', \delta(p,A) \neq \emptyset\}$$

to the set T' inductively as follows:

$$\frac{\delta(\delta(p,A),t) \neq \emptyset}{t \in \mathbf{Read}\,!\,(p,A)}\ \mathrm{DR} \qquad \frac{\delta(p,A) = r \quad C \Rightarrow^*_{P'} \varepsilon}{\mathbf{Read}\,!\,(r,C) \subseteq \mathbf{Read}\,!\,(p,A)}\ \mathrm{reads}$$

$$\frac{t \in \mathbf{Read}\,!\,(p,A)}{t \in \mathbf{Follow}\,!\,(p,A)}\ \mathrm{Read} \qquad \frac{[B \to \beta \cdot A\gamma] \in p \quad \beta : p' \to p \quad \gamma \Rightarrow^*_{P'} \varepsilon}{\mathbf{Follow}\,!\,(p',B) \subseteq \mathbf{Follow}\,!\,(p,A)}\ \mathrm{includes}$$

Then, whenever $q \in Q$ and $[A \to \omega \cdot \varepsilon] \in q$, we can compute $\mathbf{LA}(q,[A \to \omega])$ by

$$\mathbf{LA}(q,[A \to \omega]) = \{t \in T' \mid p \in Q, \omega : p \to q, \delta(p,A) \neq \emptyset, t \in \mathbf{Follow}\,!\,(p,A)\}. \tag{2}$$

Proof. Let

$$D = \{(p,A) \mid p \in Q,\ A \in N',\ \delta(p,A) \neq \emptyset\}.$$

For the proof, define two semantic sets on D. First, let

$$\mathsf{Read}(p,A) = \{t \in T' \mid \delta(p,A) = r,\ \gamma \Rightarrow^*_{P'} \varepsilon,\ \gamma t : r \to s \text{ for some } \gamma \in N'^*,\ s \in Q\}.$$

Thus $t \in \mathsf{Read}(p,A)$ exactly when, after the transition on A, one can reach a terminal transition on t through zero or more nullable nonterminal transitions. Second, let

$$\mathsf{Follow}(p,A) = \left\{t \in T' \;\middle|\; S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha A t z,\ \alpha : q_0 \to p,\ \alpha \in V'^*,\ z \in T'^*\right\}.$$

This is the semantic follow set of the nonterminal transition (p,A).

Computation of Read. We prove $\mathbf{Read}\,!\,(p,A) = \mathsf{Read}(p,A)$ on D by showing both are the least fixed point of $\{\mathrm{DR}, \mathrm{reads}\}$. For $t \in \mathsf{Read}(p,A)$ fix a witness $\gamma \in N'^*$ with $\gamma \Rightarrow^*_{P'} \varepsilon$ and $\gamma t : \delta(p,A) \to s$, and induct on $|\gamma|$:

$$\begin{aligned} |\gamma| = 0: &\quad t : \delta(p,A) \to s \overset{1(\mathrm{ix})}{\iff} s = \delta(\delta(p,A),t) \neq \emptyset \iff t \in \mathrm{DR}(p,A); \\ \gamma = C\gamma'\ (C \in N'): &\quad C \Rightarrow^*_{P'} \varepsilon,\ \gamma' \Rightarrow^*_{P'} \varepsilon,\ \gamma' t : \delta(\delta(p,A),C) \to s, \end{aligned}$$

so in the second case, with $r = \delta(p,A)$, we have $(r,C) \in D$, $(p,A) \rightsquigarrow_{\mathrm{reads}} (r,C)$, and $t \in \mathsf{Read}(r,C)$ via the shorter witness γ'; the induction hypothesis gives $t \in \mathbf{Read}\,!\,(r,C)$, whence reads yields $t \in \mathbf{Read}\,!\,(p,A)$. The reverse inclusion reads the same decomposition forwards (DR realizes $\gamma = \varepsilon$; each reads edge prepends one nullable $C \in N'$). As D, T' are finite and both clauses are monotone, the two least fixed points coincide:

$$\mathbf{Read}\,!\,(p,A) = \mathsf{Read}(p,A) \qquad ((p,A) \in D),$$

with $\mathbf{Read}\,!\,(r,C) = \emptyset$ when $(r,C) \notin D$, making the reads clause vacuous there.

Computation of Follow. We show Follow is the least solution of the clauses Read and includes, whence $\mathbf{Follow}! = \mathsf{Follow}$ on D.

Soundness of the clauses for Follow.

(Read) $\mathsf{Read}(p,A) \subseteq \mathsf{Follow}(p,A)$. Take $t \in \mathsf{Read}(p,A)$ with witness $\gamma \in N'^*$, $\gamma \Rightarrow^*_{P'} \varepsilon$, $\gamma t : \delta(p,A) \to s$. Choose $\alpha : q_0 \to p$ (possible since $p \in Q$ is accessible); then $\alpha A \gamma t : q_0 \to s$ is a viable prefix, so some rightmost derivation reaches $\alpha A \gamma t z$ ($z \in T'^*$). As γ stands immediately left of the terminals tz, it is rightmost-erasable, giving

$$S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha A \gamma t z \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha A t z, \qquad \alpha : q_0 \to p,$$

i.e. $t \in \mathsf{Follow}(p,A)$.

(includes) If $[B \to \beta \cdot A\gamma] \in p$, $\beta : p' \to p$, $\gamma \Rightarrow^*_{P'} \varepsilon$, then $\mathsf{Follow}(p', B) \subseteq \mathsf{Follow}(p, A)$. For $t \in \mathsf{Follow}(p', B)$, say $S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha' Btz$ with $\alpha' : q_0 \to p'$, expand B and erase γ rightmost:

$$\alpha' Btz \underset{\mathbf{rm}}{\Longrightarrow}_{P'} \alpha' \beta A\gamma tz \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha' \beta Atz, \qquad \alpha'\beta : q_0 \to p,$$

where $\alpha'\beta : q_0 \to p$ follows from $\alpha' : q_0 \to p'$ and $\beta : p' \to p$. With $\alpha := \alpha'\beta$ this gives $t \in \mathsf{Follow}(p, A)$.

Minimality. Let $t \in \mathsf{Follow}(p, A)$, witnessed by $S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha Atz$, $\alpha : q_0 \to p$. Let $[B \to \beta A\gamma]$ be the production creating this occurrence of A, so the derivation factors through

$$S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha' Bt'z' \underset{\mathbf{rm}}{\Longrightarrow}_{P'} \alpha'\beta A\gamma t'z' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha Atz, \quad \alpha = \alpha'\beta,\ [B \to \beta \cdot A\gamma] \in p,\ \beta : p' \to p.$$

Splitting on the residual γ:

$$\begin{cases} \gamma = \gamma_0\, t \cdots,\ \gamma_0 \Rightarrow^*_{P'} \varepsilon & \Rightarrow\ \gamma_0 t : \delta(p, A) \to (\cdot),\ \text{so } t \in \mathsf{Read}(p, A) \ \ \text{(Read)}, \\ \gamma \Rightarrow^*_{P'} \varepsilon & \Rightarrow\ t \in \mathsf{Follow}(p', B) \text{ via } (p, A) \rightsquigarrow_{\text{includes}} (p', B). \end{cases}$$

The second case strictly shortens the witnessing rightmost derivation, so iterating it terminates in the first case after finitely many steps; thus every $t \in \mathsf{Follow}(p, A)$ is generated by the clauses. Hence Follow is their least solution and

$$\mathbf{Follow}\,!\,(p, A) = \mathsf{Follow}(p, A) \qquad ((p, A) \in D).$$

It remains to translate this semantic follow set into the lookahead set of a completed item. Let $q \in Q$ and $[A \to \omega \cdot \varepsilon] \in q$. Then

$$\begin{aligned} & t \in \mathbf{LA}(q, [A \to \omega]) \\ & \quad \Longleftrightarrow \exists \alpha, z\ .S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha Atz \wedge \alpha\omega : q_0 \to q \\ & \quad \Longleftrightarrow \exists p \in Q, \omega : p \to q \wedge \delta(p, A) \neq \emptyset \wedge t \in \mathsf{Follow}(p, A) \\ & \quad \Longleftrightarrow \exists p \in Q, \omega : p \to q \wedge \delta(p, A) \neq \emptyset \wedge t \in \mathbf{Follow}\,!\,(p, A). \end{aligned}$$

The middle equivalence just cuts the path $\alpha\omega : q_0 \to q$ at the point p reached after α; conversely, concatenating $\alpha : q_0 \to p$ and $\omega : p \to q$ recovers $\alpha\omega : q_0 \to q$. The side condition $\delta(p, A) \neq \emptyset$ on the second line is automatic when $A \neq S'$: from $\omega : p \to q$ and $[A \to \omega \cdot \varepsilon] \in q$ one traces the dot back along ω to obtain $[A \to \varepsilon \cdot \omega] \in p$, and this closure item is licensed by some $[B \to \beta \cdot A\delta] \in p$, whence $\delta(p, A) \neq \emptyset$. For $A = S'$ both sides are empty: no item has a dot before S', so $\delta(p, S') = \emptyset$ for every p, and no rightmost derivation has the form $S' \underset{\mathbf{rm}}{\Longrightarrow}^*_{P'} \alpha S'tz$ since S' occurs on no right-hand side; this matches the parser accepting by reaching $(S\$: q_0 \to q_f, \varepsilon)$ rather than by reducing $[S' \to S\$]$. This is precisely the formula claimed in the theorem. □

Appendix A. Digraph algorithm and implementation specification

This appendix gives an executable specification for computing the sets used in Theorem 4. The construction is the usual digraph view of LALR(1) lookahead computation: direct seed sets are placed on nodes, dependency edges describe set inclusions, and the least fixed point is obtained by collapsing strongly connected components and propagating set unions along the resulting DAG [1, 2, 3].

A.1. General digraph fixed-point problem

Let X be a finite set of nodes, let $E \subseteq X \times X$ be a directed edge relation, and let $\mathrm{Seed} : X \to \mathcal{P}(T')$. We use the edge orientation

$$x \rightsquigarrow y \quad \text{to mean} \quad F(y) \subseteq F(x).$$

Thus the desired solution is the least function $F : X \to \mathcal{P}(T')$ satisfying

$$F(x) = \mathrm{Seed}(x) \cup \bigcup_{x \rightsquigarrow y} F(y). \tag{DG}$$

Algorithm DIGRAPH(X, E, Seed)**.**

1. Compute the strongly connected components of (X, E), for example by Tarjan's depth-first-search algorithm.
2. Build the condensation DAG $\mathcal{C}$. Its nodes are SCCs, and $C \rightsquigarrow D$ is an edge when some $x \in C$ and $y \in D$ satisfy $x \rightsquigarrow y$ and $C \neq D$.
3. Initialize
$$\mathrm{Val}(C) \leftarrow \bigcup_{x \in C} \mathrm{Seed}(x)$$
for every SCC C.
4. Process the SCCs in reverse topological order with respect to $\rightsquigarrow$, so that every successor D of C has already been processed, and put
$$\mathrm{Val}(C) \leftarrow \mathrm{Val}(C) \cup \bigcup_{C \rightsquigarrow D} \mathrm{Val}(D).$$
5. Return $F(x) := \mathrm{Val}(C_x)$, where C_x is the SCC containing x.

Correctness condition. For every SCC C, after step 4,
$$\mathrm{Val}(C) = \bigcup \{\mathrm{Seed}(y) \mid \text{some } x \in C \text{ reaches } y \text{ in } (X, E)\}.$$
Therefore the returned F is exactly the least solution of (DG). This is the only property of the graph algorithm needed by Theorem 4.

Complexity. Let $b = \lceil |T'|/w \rceil$, where w is the machine word size used for bitsets. SCC construction is $O(|X|+|E|)$, and the set propagation is $O((|X|+|E|)b)$. A simple work-list implementation that repeatedly applies $F(x) := F(x) \cup F(y)$ for $x \rightsquigarrow y$ is also correct, but the SCC version avoids repeated propagation inside cycles.

A.2. Instance for Read

Let
$$D = \{(p, A) \mid p \in Q,\ A \in N',\ \delta(p, A) \neq \emptyset\}.$$
First compute nullable nonterminals by the least fixed point
$$\mathrm{Null}(A) \iff \exists [A \to X_1 \cdots X_k] \in P' \text{ such that every } X_i \text{ is nullable},$$
where terminals are never nullable and $k = 0$ is allowed. Extend this to strings by $\mathrm{Null}(X_1 \cdots X_k)$ iff every X_i is nullable.

For each node $x = (p, A) \in D$, define the direct-read seed
$$\mathrm{DR}(p, A) = \{t \in T' \mid \delta(\delta(p, A), t) \neq \emptyset\}.$$
Define the reads dependency graph $E_R \subseteq D \times D$ by
$$(p, A) \rightsquigarrow_R (r, C) \quad \iff \quad \delta(p, A) = r \wedge C \in N' \wedge \mathrm{Null}(C) \wedge (r, C) \in D.$$
The last condition merely makes explicit the domain of Theorem 4. If $(r, C) \notin D$, then $\textbf{Read}\,!\,(r, C) = \emptyset$, so no edge is needed.

Then compute
$$\textbf{Read}\,!\,(p, A) := \mathrm{DIGRAPH}(D, E_R, \mathrm{DR})(p, A).$$
This is the least relation satisfying the DR and reads rules of Theorem 4.

A.3. Instance for Follow

The seed for follow propagation is the already computed read set:

$$\mathrm{Seed}_F(p, A) := \mathbf{Read}\,!\,(p, A).$$

Define the includes dependency graph $E_I \subseteq D \times D$ by

$$(p, A) \rightsquigarrow_I (p', B) \quad \Longleftrightarrow \quad [B \to \beta \cdot A\gamma] \in p \wedge \beta : p' \to p \wedge \mathrm{Null}(\gamma) \wedge (p', B) \in D.$$

The edge orientation again means that the right-hand node contributes to the left-hand node:

$$\mathbf{Follow}\,!\,(p', B) \subseteq \mathbf{Follow}\,!\,(p, A).$$

Then compute

$$\mathbf{Follow}\,!\,(p, A) := \mathrm{DIGRAPH}(D, E_I, \mathrm{Seed}_F)(p, A).$$

This is the least relation satisfying the Read and includes rules of Theorem 4.

A.4. Lookback relation and lookahead construction

For every completed item $[A \to \omega \cdot \varepsilon] \in q$, define its lookback set by

$$\mathrm{LB}(q, A \to \omega) := \{(p, A) \in D \mid \omega : p \to q\}.$$

The lookahead set is then computed by

$$\mathbf{LA}(q, [A \to \omega]) := \bigcup_{(p,A) \in \mathrm{LB}(q, A \to \omega)} \mathbf{Follow}\,!\,(p, A).$$

This is the formula of Theorem 4 written as an implementation step.

Implementation note. The query $\omega : p \to q$ can be implemented by following δ-edges from p along ω. The query $\beta : p' \to p$ in E_I can be implemented by caching reverse-path queries keyed by (β, p), or by storing predecessor edges during construction of the LR(0) automaton.

A.5. Parser-table and conflict specification

After **LA** has been computed, construct actions as follows.

1. If $\delta(q, t) \neq \emptyset$ for $t \in T'$, add the shift action $q \xrightarrow{t} \delta(q, t)$.
2. For each $[A \to \omega \cdot \varepsilon] \in q$ and each $t \in \mathbf{LA}(q, [A \to \omega])$, add the reduce action $A \to \omega$ under lookahead t.
3. Report a shift/reduce conflict when both a shift action and a reduce action are present for the same pair (q, t).
4. Report a reduce/reduce conflict when two distinct reduce productions are present for the same pair (q, t).

The accepting condition remains the one in Definition 1: the parser accepts when it reaches $(S\$: q_0 \to q_f, \varepsilon)$. Equivalently, in a conventional ACTION/GOTO table, q_f may be treated as the accepting state after the input marker has been consumed.

A.6. Summary specification

The complete computation required by Theorem 4 is:

1. Build G', Q, and δ as in Definition 1.
2. Compute nullable nonterminals and nullable strings.
3. Build D, DR, and E_R; compute **Read** by DIGRAPH(D, E_R, DR).
4. Build E_I; compute **Follow** by DIGRAPH$(D, E_I, \mathbf{Read})$.
5. Build LB and compute each $\mathbf{LA}(q, [A \to \omega])$ by unioning the appropriate **Follow**-sets.
6. Construct the guarded reduce actions and check conflicts.

The postcondition is that the computed **Read**, **Follow**, and **LA** are the least sets satisfying Theorem 4, and hence the lookahead-restricted parser of Theorem 3 uses exactly the LALR(1) lookahead sets determined by the LR(0) automaton.

References

[1] F. DeRemer and T. J. Pennello. Efficient computation of LALR(1) look-ahead sets. *ACM Transactions on Programming Languages and Systems*, 4(4):615–649, 1982.

[2] R. E. Tarjan. Depth-first search and linear graph algorithms. *SIAM Journal on Computing*, 1(2):146–160, 1972.

[3] J. H. Gallier. A survey of LR-parsing methods: The graph method for computing fixed points and LALR(1) lookahead sets. CIS 511 lecture notes, University of Pennsylvania, 2008.